



SOMAIYA
VIDYAVIHAR UNIVERSITY

22.05.2026 (M)

Semester: January 2026 – April 2026 (KT)

Maximum Marks: 100

Examination: ESE Examination

Duration: 3 Hrs.

Programme code: 54

Class: LY

Semester: VII (SVU 2020)

Programme: Honor in Data Science & Analytics

Institute/School/Department: K. J. Somaiya School of Engineering

Name of the department:
COMP

Course Code: 116h54C701

Name of the Course: Advanced Machine Learning

Instructions: 1) Draw neat diagrams 2) All questions are compulsory
3) Assume suitable data wherever necessary

Que. No.	Question	Max. Marks
	Solve any Four	20
i)	Briefly explain the three types of machine learning with a suitable example for each.	5
ii)	Explain overfitting and underfitting with the help of diagrams showing training versus validation loss curves.	5
iii)	Differentiate between shallow neural networks and deep neural networks, highlighting differences in architecture.	5
iv)	Explain the types of pooling operations used in Convolutional Neural Networks (CNNs).	5
v)	Explain the formation of N-gram language models with suitable examples.	5
vi)	With the aid of a diagram, explain how a pre-trained VGG16 can be fine-tuned for classifying medical X-ray images.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Consider a neuron with inputs $x_1 = 1$, $x_2 = 2$, weights $w_1 = 0.4$, $w_2 = 0.6$ and bias $= 0.5$ Apply the Sigmoid activation function and compute the output.	5
ii)	Explain the architecture of a feedforward neural network and provide the mathematical representation of each layer.	5
OR		
Q2 A	Explain how the Dropout technique improves the generalization performance of neural networks. Justify with a suitable example.	10
Q2 B	Solve any One	10
i)	Explain the working steps of the Batch Normalization algorithm.	10
ii)	Discuss the following regularization techniques with their mathematical representations: L1 Regularization (Lasso) and L2 Regularization (Ridge).	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Explain the working of convolutional layers in extracting spatial features from images. Use the example of animal classification (e.g., cat, dog, horse) to illustrate how feature maps are generated.	10
ii)	Briefly describe the architecture of a simple Recurrent Neural Network (RNN) and explain how it processes sequential data.	10
iii)	Explain the architecture of an LSTM (Long Short-Term Memory) network. Describe the role of each gate with help of suitable example.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Consider the following sentences: 1. "Students read books in the library" 2. "Teachers write notes on the board" (a) Construct the TF-IDF representation for these sentences. (b) Discuss the limitations of the TF-IDF method.	10
ii)	Discuss the CBOW and Skip-Gram architectures in the Word2Vec model with the help of the following sentence: "The dog plays in the park".	10
iii)	Explain the major components of a Natural Language Processing (NLP) workflow with help of suitable example	10

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	Compare content-based filtering and collaborative filtering approaches used in recommendation systems	5
ii)	Explain the role of learning rate in neural network optimization. Describe the effect of a very high and a very low learning rate.	5
iii)	Explain the workflow of building a customer churn prediction model using neural networks.	5
iv)	What is the Encoder-Decoder (Seq2Seq) model? Explain its architecture.	5
v)	Write a short note on Gradient Descent algorithm.	5
vi)	Explain sentiment analysis and its role in analyzing public opinion from news articles.	5